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6.1 PARASITISM AND PARASITIC ADAPTATIONS

Every living organism in this universe has an instinct to survive, for which, they require two basic needs - food and shelter. You know pretty well that green plants can prepare their own food materials in their body through photosynthesis, using CO_2 and H_2O as raw materials (**autotrophs**). But animals cannot prepare their own food materials in their body. Hence they must depend on plants or other animals for their survival (**heterotrophs**).

Certain animals are described as parasites as they survive at the cost of the other animals, called hosts. In such an association, the hosts tend to reject or resist the parasites. Concurrently the parasites have to evolve mechanisms to counteract and neutralize the host's defence in order to be successful within the host. For this purpose, the parasites have developed many special adaptations such as the loss of unnecessary sensory organs, formation of organs for adhesion, high reproductive capacity, etc.

6.1.1 Parasitism

An intimate association between two organisms of different species in which, 'one is benefited and the other one is often adversely affected' is called **parasitism**. The word parasitism comes from a Greek word '*parasitos*' (*para* - at the side of; *sitos* - food or grain) which means '*one eating at another one's table*'. The organism that obtains nourishment is called the '**parasite**' (the gainer) and the organism from which the nourishment is obtained is called the '**host**' (the loser).

I. Types of parasites

Based on the interaction between the host and the parasite, various types of parasites are recognised. Some of them are listed below.

- i) **Ectoparasite:** A parasite that lives on the surface of the host's body is called ectoparasite.

e.g. head lice and itch mites on humans, ticks on dogs, copepods on marine fishes, etc.

The female mosquito is not considered an obligate parasite though it needs human blood for fertility. Can you explain why?

Discuss with your teacher about the food habits of mosquitoes.

- ii) **Endoparasite:** A parasite that lives inside the body of the host is called endoparasite. Based on the place where they live, they are again classified into three types, namely:

- a) **Cytozoic (intracellular) parasites:** They live within the host's cells.

e.g. *Plasmodium vivax* in man, *Nosema notabilis* in *Sphaerospora polymorpha*, etc.

- b) **Histozoic (intercellular) parasites:** They live in between the cells of the tissues and organs of the host.

e.g. *Wuchereria bancrofti*, *Entamoeba histolytica*, etc.

- c) **Coelozoic parasites:** They live within the cavities of the host's body. They are called enterozoic if they live in the alimentary canal.

e.g. *Ascaris lumbricoides*

- iii) **Hyperparasite (Parasite in/on a parasite):** It is a parasite which lives in/on the body of another parasite.

e.g. *Nosema notabilis* (a cnidosporan parasite) lives in *Sphaerospora polymorpha* (also a cnidosporan parasite) which lives in the urinary bladder of toad fish.

- iv) **Monogenetic parasite:** It is a parasite which completes its life cycle in only one host.

e.g. *Entamoeba histolytica*, *Ascaris lumbricoides*, etc.

- v) **Digenetic parasite:** It is a parasite which requires at least two hosts to complete its life cycle.

e.g. *Plasmodium vivax*, *Wuchereria bancrofti*, etc.

NOTE: Parasites living in the gut of the host are called 'enterozoic parasites'. *Ascaris* is **enterozoic** and **coelozoic**, where-as *Entamoeba* is **enterozoic** and **histozoic**.

II. Types of Hosts

- i) **Primary host or Definitive host:** It is the host that harbours the adult stage or sexually mature stage of a parasite or the host in which the parasite undergoes sexual reproduction.

e.g. Man for *Wuchereria bancrofti*, female *Anopheles* for *Plasmodium*, etc.

- ii) **Intermediate host or Secondary host:** It is the host that harbours the developing 'larval or immature or asexual' stages of a parasite or the host in which the parasite undergoes asexual reproduction.

e.g. Man for *Plasmodium*, female *Culex* for *Wuchereria*, etc.

iii) Reservoir host: It is the host that lodges the infective stages of a parasite in its body when the human host is not available. In the reservoir host, the parasite neither undergoes development nor causes any disease.

e.g. Monkey for *Plasmodium*, African antelope (*Gnu*) for *Trypanosoma gambiense*, etc.

III. Vector

It is an organism (generally an insect) which transfers the infective stages of a parasite from one main host to another. Vectors are of two types, namely:

i) Mechanical vector: It is the vector, which merely transfers the infective stages of a parasite but no part of the parasitic development takes place in it.

e.g. House flies and cockroaches in the case of *Entamoeba*.

ii) Biological vector: It is the vector in which the parasite undergoes a part of the development before it gets transferred to another host.

e.g. Female *Anopheles* mosquito in the case of *Plasmodium* and female *Culex* mosquito in the case of *Wuchereria*.

6.1.2 Parasitic adaptations

Parasites have evolved special adaptations to meet the requirements and lead successful life in the hosts.

- ❖ In order to live in the host, some parasites have developed structures like hooks, suckers, rostellum, etc., for '**anchoring**'. **e.g.** *Taenia solium*
- ❖ Some intestinal parasites have developed **protective cuticle** to withstand the action of the digestive enzymes of the host. **e.g.** *Ascaris lumbricoides*
- ❖ Some intestinal parasites produce '**anti enzymes**' to neutralize the effect of host's digestive enzymes. **e.g.** *Taenia solium*
- ❖ Some parasites live as '**obligatory anaerobes**' as the availability of oxygen is very rare for them. **e.g.** *Entamoeba histolytica*, *Taenia solium*, etc.
- ❖ Some intestinal parasites live as '**facultative anaerobes**' i.e., if oxygen is not available, they live anaerobically and if oxygen is available, they respire aerobically. **e.g.** *Ascaris lumbricoides*
- ❖ The morphological and anatomical features are greatly simplified while emphasizing their **reproductive potential**. For example, an *Ascaris* lays nearly two lakh eggs per day. In *Taenia solium* the body is divided into 700 to 900 proglottids of which each proglottid acts as a unit of reproductive system and releases approximately 35,000 eggs.
- ❖ The life cycles of endoparasites are more complex because of their extreme specialization. For example, life cycle of certain parasites like *Fasciola hepatica* (sheep liver fluke) is very complex involving many developmental stages and

intermediate hosts, to increase the chances of reaching a new definitive host.

- ❖ Certain parasites like *Entamoeba* develop cysts to tide over the unfavourable conditions like desiccation while reaching the new host.
- ❖ Some parasites elude production of vaccines against them (smart parasites!) as they keep changing their surface antigens form time to time.
e.g. *Plasmodium*, HIV, etc.

Do you know: In villages, the eggs of ducks are mixed with those of hen and allowed to be incubated by hen- a kind of human induced 'brood parasitism'!

6.1.3 Effects of parasites on hosts

In general, the parasites cause weakening of the body of their hosts by causing the deprivation of nutrients, fluids and metabolites as they compete with their hosts for the same. They may also cause pathological effects in their hosts such as

- ❖ **Parasitic castration:** Some parasites cause the degeneration of gonads of the host, making it sterile. This effect is called **parasitic castration**.
e.g. *Sacculina* (root headed barnacle, a crustacean) causes the degeneration of gonads in the crab *Carcinus maenas*.
- ❖ **Neoplasia:** Some cause an abnormal growth of the host cells in a tissue to form new structures. This effect is called **neoplasia** which leads to cancers.
e.g. Some viruses
- ❖ **Gigantism:** Some parasites cause an abnormal increase in the size of the host. This effect is called **gigantism**.
e.g. The larval stages of *Fasciola hepatica* cause gigantism in snail (an intermediate host)
- ❖ **Hyperplasia:** Some parasites cause increase in the **number of cells**. This effect is called **hyperplasia**.
e.g. *Fasciola hepatica* in the bile duct of sheep
- ❖ **Hypertrophy:** Some parasites cause an abnormal increase in the **volume/size** of the infected host cells. This effect is called **hypertrophy**.
e.g. RBC of man infected by *Plasmodium*
- ❖ Most of the parasites cause various types of diseases like
 - i) African sleeping sickness by *Trypanosoma gambiense*
 - ii) Delhi boils/Tashkent ulcers/Oriental sores by *Leishmania tropica*
 - iii) Kala azar/Dum dum fever/Visceral leishmaniasis by *Leishmania donovani*
 - iv) Malaria by *Plasmodium* sps
 - v) Elephantiasis by *Wuchereria bancrofti*.

'An ideal parasite should be able to thrive within the host without harming it'. Do you believe in it? Then why didn't Natural Selection lead to the evolution of such totally harmless parasites? Clue: Think of the phenomenon of 'Ecological balance in nature'